

MIPT & YSDA · AUTUMN 2026

# Deep Generative Models

## Lecture 8

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# Recap of Previous Lecture

## Forward Gaussian Diffusion Process

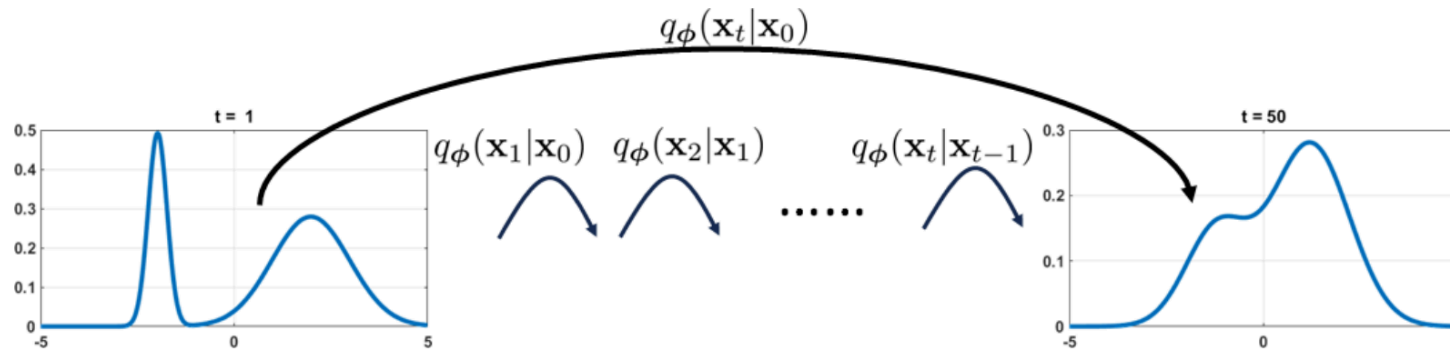
Let  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ,  $\beta_t \ll 1$ ,  $\alpha_t = 1 - \beta_t$  and  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ .

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \boldsymbol{\epsilon}_t, \quad \text{where } \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I});$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

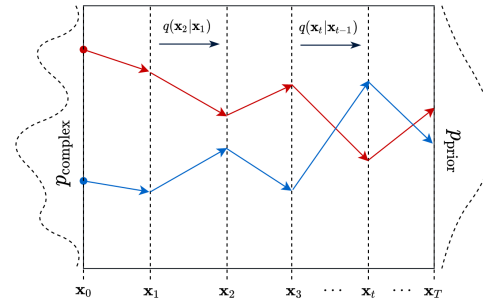
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I});$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}).$$



# Recap of Previous Lecture

**Diffusion** describes the migration of particles from regions of high density to those of low density.



1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}), t \geq 1$
3. After  $T \gg 1$  steps:  $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

If this process can be reversed, we can sample from  $p_{\text{data}}(\mathbf{x})$  by starting from noise  $p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ . Our goal now becomes inverting this diffusion.

# Recap of Previous Lecture

## NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

# Recap of Previous Lecture

## NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

## Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0}{1 - \bar{\alpha}_t}$$

# Recap of Previous Lecture

## Theorem (Denoising Score Matching)

$$\begin{aligned} \mathbb{E}_{q(\mathbf{x}_t)} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)\|_2^2 \\ = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta}) \end{aligned}$$

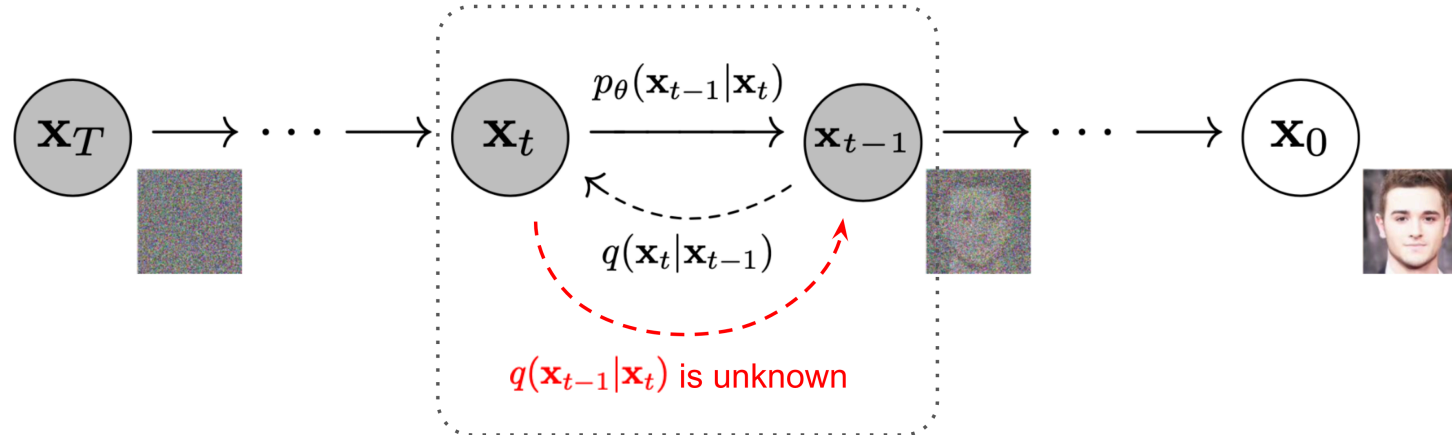
# Recap of Previous Lecture

## Theorem (Denoising Score Matching)

$$\begin{aligned} \mathbb{E}_{q(\mathbf{x}_t)} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)\|_2^2 \\ = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \|\mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x})\|_2^2 + \text{const}(\boldsymbol{\theta}) \end{aligned}$$

**Note:** Annealed Langevin dynamics applies to diffusion, too.

# Recap of Previous Lecture



## Reverse Process (Ancestral Sampling)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}$$

$$\approx p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta,t}(\mathbf{x}_t), \sigma_{\theta,t}^2(\mathbf{x}_t))$$

Feller's theorem justifies this Gaussian assumption.

# Recap of Previous Lecture

## Forward Process

1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1} + \sqrt{\beta_t} \epsilon_t$
3.  $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

## Reverse Process

1.  $\mathbf{x}_T \sim p_{\infty}(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$
2.  $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$
3.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$

# Recap of Previous Lecture

**Forward process** maps any distribution  $p_{\text{data}}(\mathbf{x})$  to  $\mathcal{N}(0, \mathbf{I})$  by injection of noise:

$$\begin{aligned} q(\mathbf{x}_t | \mathbf{x}_{t-1}) &= \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}); \\ q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \end{aligned}$$

**Reverse process** refers to an intractable distribution that can be approximated by a normal distribution (with unknown parameters) for small  $\beta_t$ :

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1}) q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)} \approx \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

**Conditioned reverse process** is a normal distribution with known parameters, describing how to denoise a noisy image  $\mathbf{x}_t$  when we know the clean image  $\mathbf{x}_0$ .

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$$

# Recap of Previous Lecture

- $\mathbf{z} = (\mathbf{x}_1, \dots, \mathbf{x}_T)$  represents the latent variables.
- Variational posterior distribution:

$$q(\mathbf{z}|\mathbf{x}) = q(\mathbf{x}_1, \dots, \mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}).$$

- Generative model and prior:

$$p_{\theta}(\mathbf{x}|\mathbf{z}) = p_{\theta}(\mathbf{x}_0|\mathbf{x}_1); \quad p_{\theta}(\mathbf{z}) = \prod_{t=2}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \cdot p(\mathbf{x}_T)$$

# Recap of Previous Lecture

## Standard ELBO

$$\begin{aligned}\log p_{\theta}(\mathbf{x}) &\geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z}|\mathbf{x})} = \mathcal{L}_{\phi, \theta}(\mathbf{x}) \rightarrow \max_{\phi, \theta} \\ \mathcal{L}_{\phi, \theta}(\mathbf{x}) &= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\ &\quad - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

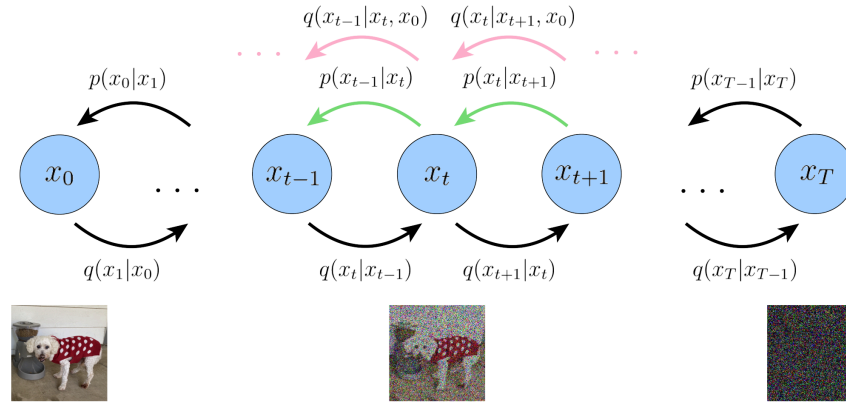
# Outline

- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization
- 03 Denoising Diffusion Probabilistic Model (DDPM)
- 04 Model Guidance
  - Classifier Guidance
  - Classifier-Free Guidance

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# ELBO for Gaussian Diffusion Model



$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\boldsymbol{\beta}}_t \mathbf{I}),$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))$$

# ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \\ p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))\end{aligned}$$

# ELBO for Gaussian Diffusion Model

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Let's assume that

$$\begin{aligned}\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t) &= \tilde{\beta}_t \mathbf{I} \\ \Rightarrow p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}).\end{aligned}$$

# ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\ q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_{t-1} | \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \\ p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1} | \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t))\end{aligned}$$

Let's assume that

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Theoretically, the optimal  $\boldsymbol{\sigma}_{\theta,t}^2(\mathbf{x}_t)$  lies in  $[\tilde{\beta}_t, \beta_t]$ :

- $\beta_t$  is optimal for  $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$ ;
- $\tilde{\beta}_t$  is optimal for  $\mathbf{x}_0 \sim \delta(\mathbf{x}_0 - \mathbf{x}^*)$ .

# ELBO for Gaussian Diffusion Model

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}} \left( \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \right. \\ \left. \parallel \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}) \right)$$

# ELBO for Gaussian Diffusion Model

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}} \left( \mathcal{N}(\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \right. \\ &\quad \left. \parallel \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t), \tilde{\beta}_t \mathbf{I}) \right) \\ &= \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]\end{aligned}$$

# ELBO for Gaussian Diffusion Model

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute noisy image  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$ .
3. Compute the ELBO:

$$\mathcal{L}_{\phi, \theta}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1 | \mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p(\mathbf{x}_T)) \\ - \underbrace{\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta, t}(\mathbf{x}_t)\|^2 \right]}_{\mathcal{L}_t}.$$

# ELBO for Gaussian Diffusion Model

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## Sampling

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise  $\mathbf{x}_{t-1} = \mu_{\theta, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

# Outline

- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization**
- 03 Denoising Diffusion Probabilistic Model (DDPM)
- 04 Model Guidance
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# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$
$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$
$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$$

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$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon} \quad \Rightarrow \quad \mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_0$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon} \quad \Rightarrow \quad \mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$

- There is a linear relationship between  $\boldsymbol{\epsilon}$ ,  $\mathbf{x}_t$ , and  $\mathbf{x}_0$ .
- Let's try to rewrite this mean using only  $\mathbf{x}_t$  and  $\boldsymbol{\epsilon}$ .

# Reparametrization of DDPM

$$\begin{aligned}\tilde{\mu}_t(\mathbf{x}_t, \epsilon) = & \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t \\ & + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left( \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon}{\sqrt{\bar{\alpha}_t}} \right)\end{aligned}$$

# Reparametrization of DDPM

$$\begin{aligned}\tilde{\mu}_t(\mathbf{x}_t, \epsilon) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \cdot \mathbf{x}_t \\ &\quad + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \cdot \left( \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon}{\sqrt{\bar{\alpha}_t}} \right) \\ &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t}(1 - \bar{\alpha}_t)} \cdot \epsilon\end{aligned}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

## Reparametrization

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) \right\|^2 \right]$$

## Reparametrization

$$\begin{aligned} \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon} \\ \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) &= \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t) \end{aligned}$$

# Reparametrization of DDPM

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_{\theta, t}(\mathbf{x}_t)\|^2 \right]$$

# Reparametrization of DDPM

$$\begin{aligned}
 \mathcal{L}_t &= \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta, t}(\mathbf{x}_t) \right\|^2 \right] \\
 &= \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta, t}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}) \right\|^2 \right]
 \end{aligned}$$

# Reparametrization of DDPM

$$\begin{aligned}\mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\mathbf{x}_t) \right\|^2 \right] \\ &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon) \right\|^2 \right]\end{aligned}$$

At every step of the reverse process, we attempt to predict the noise  $\epsilon$  that was used in the forward diffusion process!

# Reparametrization of DDPM

$$\begin{aligned}\mathcal{L}_{\phi, \theta}(\mathbf{x}) = & \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) - D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \| p(\mathbf{x}_T)) \\ & - \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{\mathcal{L}_t}\end{aligned}$$

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$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ \frac{(1 - \alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_{\theta, t}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon) \right\|^2 \right]$$

# Reparametrization of DDPM

Let's drop the scaling coefficient; the reconstruction term ( $t = 1$ ) has the same form.

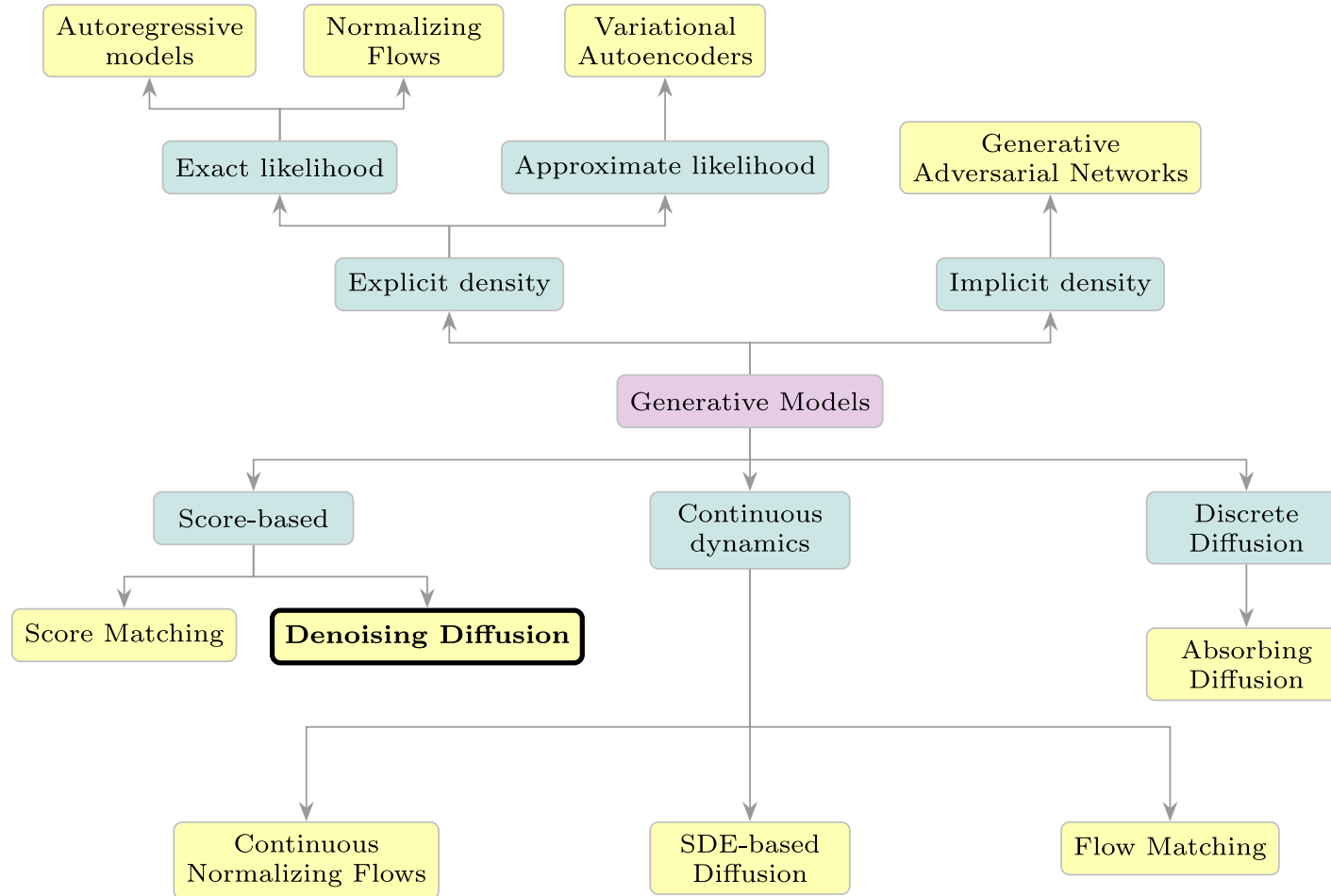
## Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left\| \epsilon - \epsilon_{\theta, t} \left( \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon \right) \right\|^2$$

# Outline

- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization
- 03 Denoising Diffusion Probabilistic Model (DDPM)**
- 04 Model Guidance
  - Classifier Guidance
  - Classifier-Free Guidance

# Generative Models Taxonomy



# Denoising Diffusion Probabilistic Model (DDPM)

## DDPM is a VAE Model

- The encoder is a fixed Gaussian Markov chain  $q(\mathbf{x}_1, \dots, \mathbf{x}_T | \mathbf{x}_0)$ .
- The latent variable is hierarchical (at each step, its dimension equals the input's).
- The decoder is a simple Gaussian model  $p_{\theta}(\mathbf{x}_0 | \mathbf{x}_1)$ .
- The prior distribution is given by a parametric Gaussian Markov chain  $p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$ .

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## Forward Process

1.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ ;
2.  $\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t$ ;
3.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ .

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3.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ .

## Reverse Process

1.  $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$ ;
2.  $\mathbf{x}_{t-1} = \sigma_{\theta,t}(\mathbf{x}_t) \cdot \epsilon + \mu_{\theta,t}(\mathbf{x}_t)$ ;
3.  $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ .

# Denoising Diffusion Probabilistic Model (DDPM)

## Training

1. Sample  $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$ ,  $t \sim U\{1, T\}$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute noisy image  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon$ .
3. Compute loss  $\mathcal{L}_{\text{simple}} = \|\epsilon - \epsilon_{\theta,t}(\mathbf{x}_t)\|^2$ .

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## Sampling (Ancestral)

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Compute the mean of  $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$ :

$$\boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \epsilon_{\theta,t}(\mathbf{x}_t).$$

3. Denoise  $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\theta,t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ .

# Denoising Diffusion as a Score-Based Generative Model

## DDPM Objective

$$\mathcal{L}_t = \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{1,t} \cdot \|\epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon\|_2^2 \right]$$

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 \mathcal{L}_t &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{1,t} \cdot \left\| \epsilon_{\theta,t}(\mathbf{x}_t) - \epsilon \right\|_2^2 \right] \\
 &= \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[ C_{2,t} \cdot \left\| \frac{\epsilon_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} - \frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \right\|_2^2 \right] \\
 q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0, (1 - \bar{\alpha}_t) \cdot \mathbf{I}) \\
 \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) &= -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0}{1 - \bar{\alpha}_t} = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}}.
 \end{aligned}$$

# Denoising Diffusion as a Score-Based Generative Model

We can reparameterize the model as:

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t) = -\frac{\boldsymbol{\epsilon}_{\theta,t}(\mathbf{x}_t)}{\sqrt{1 - \bar{\alpha}_t}} = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t).$$

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$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ C_{2,t} \cdot \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|_2^2 \right]$$

# DDPM vs NCSN: Objectives

## DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left[ C_{2,t} \left\| \mathbf{s}_{\theta,t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2 \right]$$
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In practice, this coefficient is often omitted.

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In practice, **this coefficient** is often omitted.

## NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

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$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon$$

**Maximizing the ELBO leads to the same objective as denoising score matching!**

# DDPM vs NCSN: Sampling

## Sampling (Ancestral)

$$\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$$

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

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# DDPM vs NCSN: Sampling

## Sampling (Annealed Langevin Dynamics)

1. Sample  $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$ .
2. Update  $\mathbf{x}_t^l$  via  $L$  steps of Langevin dynamics at each noise level  $\sigma_t$ :

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \cdot \mathbf{s}_{\theta, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \cdot \boldsymbol{\epsilon}_t^l.$$

3. Update  $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$  and proceed to the next noise level  $\sigma_{t-1}$ .

# DDPM vs NCSN: Summary

## Summary

- Different Markov chains:
  - DDPM:  $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \epsilon;$
  - NCSN:  $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \epsilon.$
  - One can generalize to  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$

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- The objectives coincide: ELBO  $\equiv$  score-matching.

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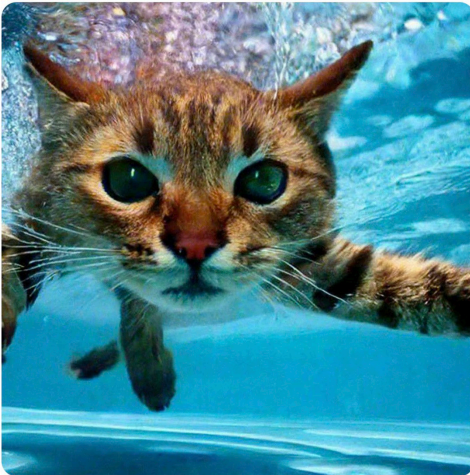
- Different Markov chains:
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  - One can generalize to  $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\alpha_t \cdot \mathbf{x}_0, \sigma_t^2 \mathbf{I}).$
- The objectives coincide: ELBO  $\equiv$  score-matching.
- The sampling procedures differ:
  - Ancestral sampling in DDPM;
  - Annealed Langevin dynamics for NCSN;
  - Hybrid approaches that combine both updates are possible.

# Outline

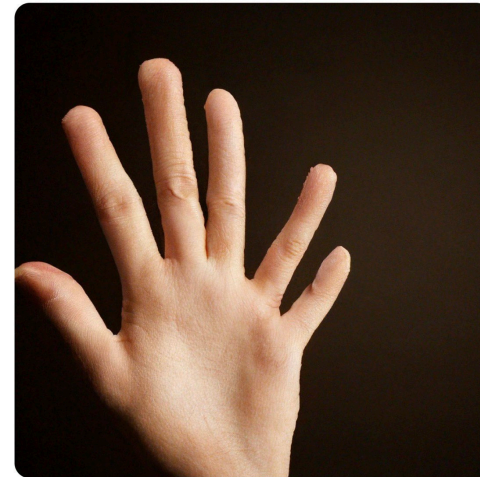
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# Guidance

- Up to now, we have focused on **unconditional** generative models  $p_{\theta}(\mathbf{x})$ .
- In practice, most generative models are **conditional** (in the diffusion era it is called guided):  $p_{\theta}(\mathbf{x}|\mathbf{y})$ .
- Here,  $\mathbf{y}$  might denote a class label or **text** (as in text-to-image tasks).



Кот ныряет в бассейн, как ребенок на обложке альбома Nevermind, реалистично



рука человека с пятью пальцами, ни четырьмя, ни шестью, а с 5 (пять) пальцами

# Conditional Models

In practice, we're typically interested in learning conditional models (sampling from conditional distribution  $p_{\text{data}}(\mathbf{x}|\mathbf{y})$ ).

- $\mathbf{y} = \emptyset, \mathbf{x} = \text{image} \Rightarrow$  unconditional image model
- $\mathbf{y} = \text{class label}, \mathbf{x} = \text{image} \Rightarrow$  class-conditional image model
- $\mathbf{y} = \text{text prompt}, \mathbf{x} = \text{image} \Rightarrow$  text-to-image model
- $\mathbf{y} = \text{image}, \mathbf{x} = \text{image} \Rightarrow$  image-to-image model
- $\mathbf{y} = \text{image}, \mathbf{x} = \text{text} \Rightarrow$  image-to-text (image captioning) model
- $\mathbf{y} = \text{English text}, \mathbf{x} = \text{Russian text} \Rightarrow$  sequence-to-sequence model (machine translation)
- $\mathbf{y} = \text{sound}, \mathbf{x} = \text{text} \Rightarrow$  speech-to-text (automatic speech recognition) model
- $\mathbf{y} = \text{text}, \mathbf{x} = \text{sound} \Rightarrow$  text-to-speech model

# Label Guidance

**Label:** Ostrich (10th ImageNet class)



**VQ-VAE (Proposed)**

**BigGAN deep**

# Text Guidance

**Prompt:** a stained glass window of a panda eating bamboo

Left:  $\gamma = 1$ , Right:  $\gamma = 3$ .



# Guidance in Generative Models

## How to make a guided model?

Instead of sampling from  $p_{\theta}(\mathbf{x})$ , we sample from  $p_{\theta}(\mathbf{x}|\mathbf{y})$ .

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- $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$  for DDPMs.

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- $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{y})$  for DDPMs.

## Challenge

- Empirically, images sampled with this procedure do not fit well enough to the desired label  $\mathbf{y}$ .
- Being able to control the strength of guidance is especially valuable.

# Outline

- 01 Diffusion ELBO Derivation (continued)
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# Classifier Guidance

## DDPM Sampling

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise the image (unconditional generation):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon$$

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## DDPM Sampling

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise the image (unconditional generation):

$$\begin{aligned}\mathbf{x}_{t-1} &= \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \sigma_t \cdot \epsilon \\ &= \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \sigma_t \cdot \epsilon\end{aligned}$$

## Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

# Classifier Guidance

## DDPM Sampling

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2. Denoise the image (unconditional generation):

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## Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

What is the link between  $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t)$  and  $\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y})$ ?

# Classifier Guidance: Guided Score Function

## Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

# Classifier Guidance: Guided Score Function

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## Guided Generation

$$\nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log \left( \frac{p_{\theta}(\mathbf{x}_t) p(\mathbf{y} | \mathbf{x}_t)}{p(\mathbf{y})} \right)$$

# Classifier Guidance: Guided Score Function

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# Classifier Guidance: Guided Score Function

## Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) + \sigma_t \cdot \epsilon$$

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# Classifier Guidance: Guided Score Function

## Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}).$$

# Classifier Guidance: Guided Score Function

## Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}).$$
$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

# Classifier Guidance: Guidance Scale

## Guided Score Function

$$\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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## Guided Score Function

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- Let us assume  $\mathbf{y}$  is a class label.
- $p(\mathbf{y}|\mathbf{x}_t)$  is a classifier for noisy inputs.
- $p(\mathbf{y}|\mathbf{x}_t)$  is responsible for model guidance.

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## Guidance Scale

It is a natural idea to scale up the contribution of the guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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- The **guidance scale**  $\gamma$  adjusts the strength of classifier guidance.
- $\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y})$  is not the true guided score function  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Classifier Guidance: Distribution Sharpening

## Scaled Guided Score Function

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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## Scaled Conditional Distribution

$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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# Classifier Guidance: Distribution Sharpening

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**Note:** Increasing  $\gamma$  sharpens  $p(\mathbf{y}|\mathbf{x}_t)$ , increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

# Classifier Guidance: Overview

## Training

1. Train the DDPM as before.
2. Train an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  on noisy data (time-dependent).

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## Sampling (Guided)

1. Sample  $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ .
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

# Outline

- 01 Diffusion ELBO Derivation (continued)
- 02 Gaussian Diffusion Reparametrization
- 03 Denoising Diffusion Probabilistic Model (DDPM)
- 04 Model Guidance**
  - Classifier Guidance
  - Classifier-Free Guidance**

# Classifier-Free Guidance

- The previous approach relies on training an additional classifier  $p(\mathbf{y}|\mathbf{x}_t)$  for noisy images.
- We now introduce a method to sidestep this requirement.

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$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

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## Bayes theorem

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log \left( \frac{p_\theta(\mathbf{x}_t|\mathbf{y})p(\mathbf{y})}{p_\theta(\mathbf{x}_t)} \right)$$

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# Classifier-Free Guidance

## Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^{\gamma} \log p_{\theta}(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_{\theta}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y} | \mathbf{x}_t)$$

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# Classifier-Free Guidance: Formulation

## Scaled Guided Score Function

$$\nabla_{\mathbf{x}_t}^\gamma \log p_\theta(\mathbf{x}_t|\mathbf{y}) = (1 - \gamma) \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t|\mathbf{y})$$

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## Naive training approach

- Train an unguided score function model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$ .
- Train a guided score function model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Classifier-Free Guidance: Formulation

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- Train an unguided score function model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t)$ .
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- Use their convex combination at inference.

## Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^\gamma(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

# Classifier-Free Guidance: Formulation

## Scaled Guided Score Function

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## Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^\gamma(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon$$

How to avoid training two separate score function models?

# Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

## CFG

1. Introduce the "absence of conditioning" label  $\mathbf{y} = \emptyset$ .
2. Identify the unguided score function  $\mathbf{s}_{\theta,t}(\mathbf{x}_t) = \mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$ .
3. Train a single model  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$  on **supervised** data, dropping the label  $\mathbf{y}$  with some fixed probability (simulating  $\mathbf{y} = \emptyset$ ).
4. At inference, evaluate the model twice to obtain  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \emptyset)$  and  $\mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$ .

# Classifier-Free Guidance

$$\mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) = (1 - \gamma) \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\theta,t}(\mathbf{x}_t, \mathbf{y})$$

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## Sampling (Guided)

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1 - \beta_t}} \cdot \mathbf{s}_{\theta,t}^{\gamma}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \epsilon.$$

# Summary

- At each step, DDPM predicts the noise that was injected in the forward process.
- DDPMs are quite slow, since the model must be applied  $T$  times for sampling.
- DDPM and NCSN are intimately connected at the objective level.
- Guidance technique allows to make controllable generation via conditioning on labels or text prompts.
- Classifier guidance turns an unconditional model into a conditional one by training an auxiliary classifier on noisy data.
- Classifier-free guidance removes the need for such a classifier, yielding a practical recipe now widely used.